

CS486 MOCK FINAL EXAMINATION - TEST 3

Based on the FINAL REVIEW structure; total: 10 points

Question 1 - ER Models

[1 pt]

An airline maintains **Flights**, **FlightLegs**, **Airports**, **Aircraft**, and dated **LegInstances**. A flight number uniquely identifies a flight. A leg is numbered 1,2,3,... within a flight. Each leg has an origin and destination airport. A leg instance is identified by flight number, leg number, and date and uses one aircraft. Airports have unique codes; aircraft have unique IDs.

Draw the ER model with weak/composite identification where appropriate and map FlightLeg and LegInstance to relations.

Question 2 - Query Languages

[2 pt]

Schemas:

$Flight(FNo, Airline)$, $Leg(FNo, LegNo, Orig, Dest)$, $Instance(FNo, LegNo, FDate, AircraftID, SeatsBooked)$

Write:

- (a) RA to find flights having at least one leg originating in SGN and at least one (possibly different) leg arriving at HAN.
- (b) TRC to find airports that are never used as a destination.
- (c) SQL to find, for every flight, the date(s) on which its total seats booked across all legs is the maximum for that flight. Keep ties.

Question 3 - Database Constraints

[1 pt]

Business rule: *For every flight leg, origin and destination airports must be different; and on a given dated leg instance, SeatsBooked cannot exceed the capacity of the assigned aircraft.*

Assume $Aircraft(AircraftID, Capacity)$. Express the first as a CHECK and the second as a formal cross-relation constraint plus trigger/assertion pseudocode.

Question 4 - Serialization & Concurrency

[2 pt]

Given

$$S = r_1(A) \ r_2(A) \ w_1(B) \ w_3(A) \ r_2(B) \ w_2(C) \ r_3(C) \ w_1(C).$$

- (a) Build the precedence graph, determine conflict serializability, and list a cycle if one exists.
- (b) If commits are c_1, c_3, c_2 , determine whether the schedule is recoverable and cascadeless, assuming reads can see previous writes.
- (c) State the growing and shrinking phases of 2PL and explain why acquiring a new lock after the first unlock violates 2PL.

Question 5 - Recovery**[1 pt]**

Under **UNDO-only** logging, the log contains old values:

```
<start T1>
<T1, A, 50>
<start T2>
<T2, B, 60>
<commit T1>
<T2, C, 70>
--- crash ---
```

Identify the winner and loser. Specify exactly which data items must be undone and why committed T1 does not need REDO under the stated protocol.

Question 6 - Functional Dependencies & Normalization**[1 pt]**

Let $R(A, B, C, D, E, G)$ and

$$F = \{A \rightarrow BC, B \rightarrow D, CD \rightarrow E, E \rightarrow G, G \rightarrow A\}.$$

- (a) Compute B^+ and A^+ .
- (b) Find at least two candidate keys, or prove there is only one.
- (c) Identify one BCNF violation, if any, and propose one lossless BCNF decomposition step.

Question 7 - Concurrency Control Execution**[1 pt]**

Two counters start at $X = 5, Y = 5$.

```
T1: r(X); r(Y); if X+Y >= 10 then X:=X-5; w(X)
T2: r(X); r(Y); if X+Y >= 10 then Y:=Y-5; w(Y)
```

Give a nonserial interleaving in which both conditions are evaluated true and state the final (X, Y) . Name the anomaly and explain what invariant is violated.

Question 8 - Query Optimization**[1 pt]**

FlightInstance has 2,400,000 tuples, 120 tuples/page (20,000 pages). A clustered B+ tree on FDate has height 3. A hash index exists on AircraftID. A query requests all rows for one month, estimated at 1/24 of the table.

Estimate B+ tree I/O as index descent plus sequential data pages, compare with full scan, and state whether the hash index can efficiently support the month range predicate.